

Technical Specification For Digital Futures Club IT Infrastructure

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Abstract

This document outlines a formal proposal for the acquisition and implementation of initial IT equipment for the newly established Digital Futures Club at NED University. The proposal details hardware specifications for two portable computing units, one stationary desktop unit, printing peripherals, and a data management strategy. The total estimated cost of \$2,400 falls within the club's allocated \$2,500 grant budget.

1 Introduction

The Digital Futures Club requires foundational IT infrastructure to support its mandate of digital literacy workshops and collaborative project development (e.g., website/app development). The specific needs identified include portable computing for administrative presentations and a dedicated, higher-specification stationary unit for general member use and development tasks.

2 Hardware Specifications

The following hardware configurations are recommended for procurement:

2.1 Portable Computing Units (Laptops, Quantity: 2)

These units are intended for use by the club president and treasurer for administrative work and presentations.

- **Model:** Dell Inspiron 14 5000 Series (Example Model - ensure you use real specs)
- **CPU:** Intel Core i5-1235U, 1.3 GHz (up to 4.4 GHz Turbo)
- **RAM:** 8 GB DDR4-3200 SDRAM
- **Storage:** 256 GB PCIe NVMe M.2 SSD
- **Display:** 14.0" Full HD (1920 x 1080) IPS Display
- **OS:** Windows 11 Home

2.2 Stationary Computing Unit (Desktop, Quantity: 1)

This unit will serve as the "club station" for collaborative project work.

- **Model:** HP Pavilion Desktop TE02 (Example Model - ensure you use real specs)
- **CPU:** AMD Ryzen 5 5600G, 3.9 GHz (up to 4.4 GHz Max Boost)
- **RAM:** 16 GB DDR4-3200 SDRAM

- **Storage:** 512 GB PCIe NVMe M.2 SSD + 1TB HDD
- **Peripherals:** Basic Keyboard and Mouse included (Monitor not included in budget)
- **OS:** Windows 11 Home

2.3 Printing Solution (Quantity: 1)

A basic color inkjet printer for flyers and documentation.

- **Model:** HP ENVY 6055e All-in-One Printer (Example Model)
- **Connectivity:** Wi-Fi, USB 2.0

3 Budget Table

The total budget allocation is detailed in Table 1, ensuring all costs are within the \$2,500 grant limit.

Table 1: Proposed Equipment Budget

Item	Model/Specs	Unit Price (\$)	Qty	Total Price (\$)
Laptop	i5, 8GB RAM, 256GB SSD	600.00	2	1200.00
Desktop PC	Ryzen 5, 16GB RAM, 512GB SSD	700.00	1	700.00
Printer	HP Envy 6055e	100.00	1	100.00
Software/Misc.	OS licenses included, miscellaneous cables	N/A	N/A	400.00
Grand Total (\$):				2400.00

4 Software and Data Management

The primary OS across all units will be Windows 11 Home (licenses bundled with hardware purchases). Key software applications include the Microsoft Office Suite for administrative documentation and financial tracking, and Visual Studio Code (VS Code), an open-source integrated development environment critical for project development workshops.

For data redundancy and integrity, a cloud-based backup solution is proposed. All critical project files, club member lists, and budget documents generated on the stationary desktop unit will be synchronized nightly to a dedicated cloud storage service (e.g., University-provided Google Drive or OneDrive account). This aims to ensure data durability and mitigates the data safety loss due to localized hardware failure, adhering to standard IT best practices for data management.